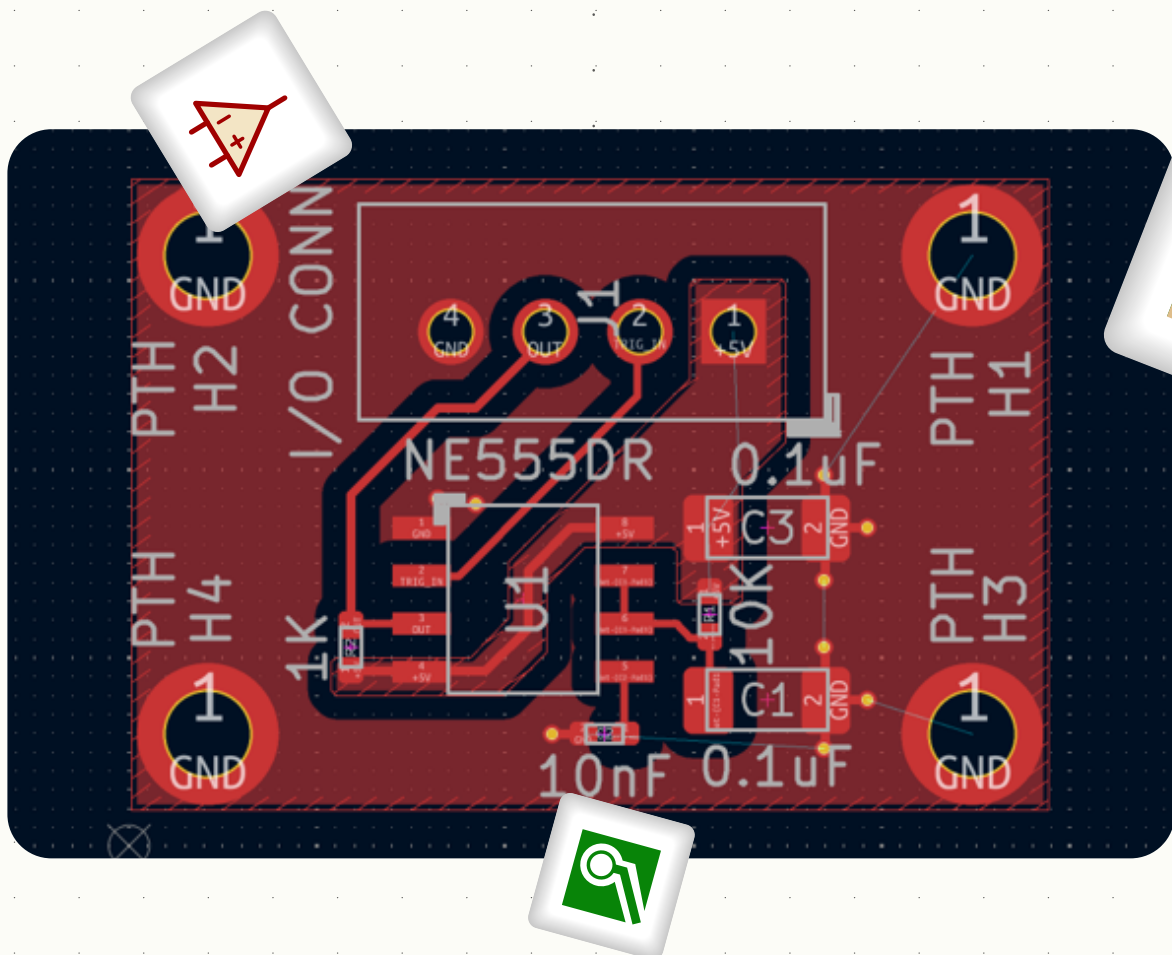


KiCad 9

Step-by-Step Guide

The beginners' guide to KiCad 9 from conception through end results



Written by **Steve Kellner**

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Preface

The content of this beginner-friendly KiCad guide was generously shared with us by Steve Kellner. As Steve describes it, his goal was to create a straightforward, step-by-step resource for newcomers who find themselves overwhelmed by the complexity of KiCad's official documentation.

In his own words, Steve started this guide as a personal learning tool, recognizing that teaching (or writing instructions) is one of the best ways to truly understand a subject. Drawing from his own early struggles, he crafted a guide that walks users through the entire process, from creating a new project to generating Gerber files, with a focus on clarity and simplicity.

We're grateful to Steve for allowing us to share his guide more broadly. We hope it proves as helpful to you as it has been for him and for others just starting their journey with KiCad.

About the Author



Steve Kellner is now retired, following a long and accomplished career in electronics and medical instrumentation. After studying electronics, he served as the chief engineer at two radio stations in the New York metropolitan area.

He also spent time in the U.S. Air Force working on ground navigation systems. For the next 50 years, Steve dedicated himself to the field of medical instrumentation, specifically electromyography (EMG), where he was involved in product development, application engineering, customer training, writing technical specifications, and product manuals.

In retirement, Steve enjoys spending time with his family, traveling, and pursuing his passion for model railroading. About 15 years ago, the model train club he belongs to asked him to design new 12-volt DC train throttles for their layout.

While his original designs worked well, they proved fragile over time. Now, as they begin to fail, Steve has proposed rebuilding them using fabricated PCBs. This project marked his introduction to the world of CAD, an experience that has proven challenging and rewarding.

1.1 An overview of KiCad PCB design tool

KiCad is an open-source software for electronic design automation (EDA) that enables engineers, hobbyists, and students to design PCBs with ease. With an active development community and professional-level features, KiCad has grown into one of the most widely used tools for printed board design.

It offers a complete set of tools for creating schematics, assigning footprints, designing layouts, and generating the manufacturing files required to produce physical boards.

The platform allows you to create custom symbol and footprint libraries, perform electrical (ERC) and design rule checks (DRC), view 3D previews, and export Gerber and drill files.

KiCad 9 brings a host of improvements to make PCB design easier and more efficient. Key updates include a new Zone Manager, better net and rule checking, and the ability to set do not populate (DNP) options at the sheet level.

It also adds helpful usability features like selection filters, text alignment shortcuts, via tenting control, and improved 3D model exports.

This guide is intended for those of you who are new to CAD and/or KiCad 9.

Success comes from good planning!

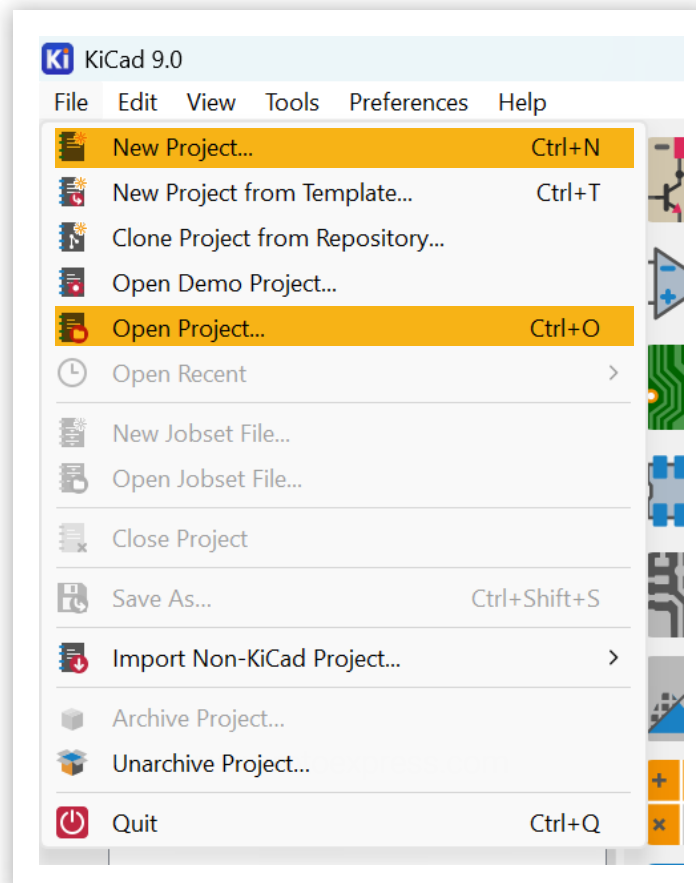
1.2 KiCad project workflow overview

Although there are many ways to create and navigate through a project, the sequence below should save you time and effort. It also helps prevent errors.

1. Start a new project or **open** an existing project.
2. Open the symbol editor in the KiCad window by clicking on **Tools > Symbol Editor**.
3. Create one or more symbol libraries.
4. Populate your new symbol library or libraries.
5. Open the footprint editor in the KiCad window by selecting **Tools > Footprint Editor**.
6. Create one or more footprint libraries.
7. Populate your new footprint library or libraries.
8. Fill in your symbol properties. This step includes linking footprints to symbols.
9. Open the schematic editor by clicking **Tools > Schematic Editor**.
10. Create a schematic. In this step, additional symbols are added through symbol duplication.
11. Run the electrical rules checker to fix the errors, if any.
12. Open the PCB editor by clicking on **Tools > Switch to PCB Editor**.
13. Update the PCB editor from the schematic by clicking **Tools > Update PCB from Schematic**.
14. Rearrange the footprints as necessary in the PCB editor.
15. Run the design rules checker and fix the errors, if any.
16. Create Gerber files by clicking on **File > Fabricate Outputs > Gerber**.

Creating a new project, symbol, and footprint library

02



Starting a new project or opening an existing project

2.1 Starting a new project

1. Verify if KiCad is pointing to the correct Windows directory.
2. Open KiCad.
3. Select **File** > **New Project**.
4. In the **Create New Project** window, select the location for your Windows directory (e.g. D:\KiCad Projects).

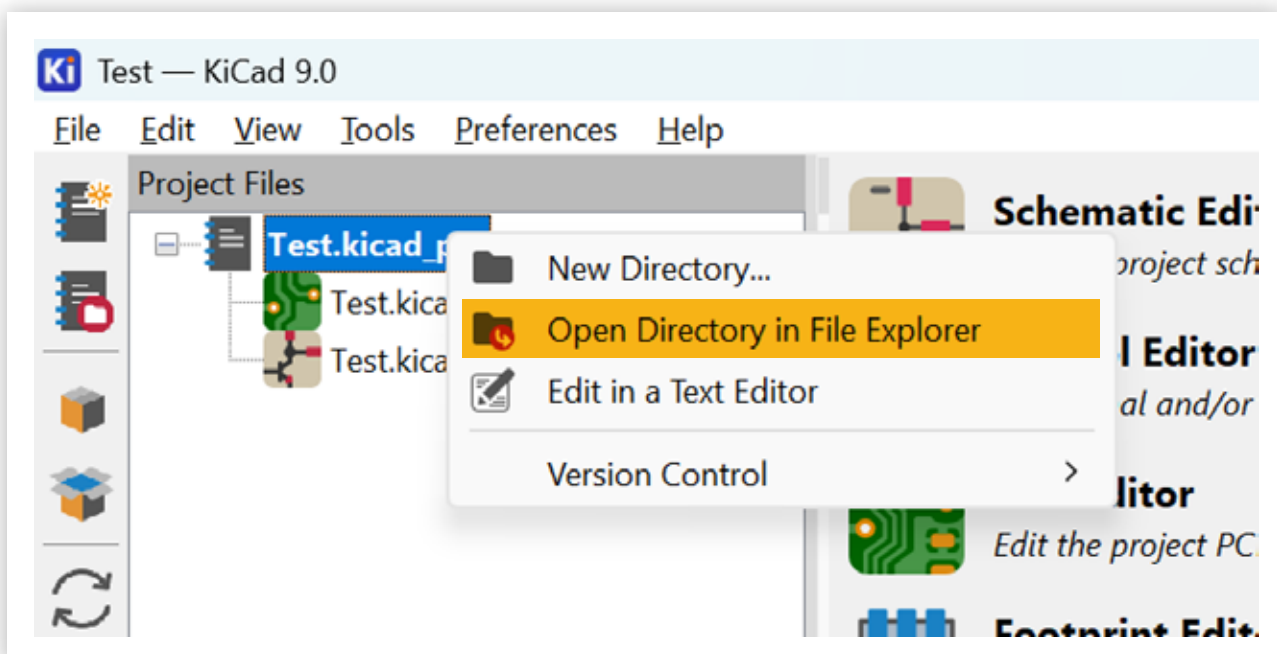
5. Enter the project's name. KiCad will add a file extension named **".kicad_pro"**
6. Click **Save**.

2.2 Opening an existing project

1. Open **KiCad**.
2. Select **File > Open Project**.
3. Right-click on the project you'd like to open.
4. Select **Open**.

2.3 Creating folders for symbol and footprint libraries

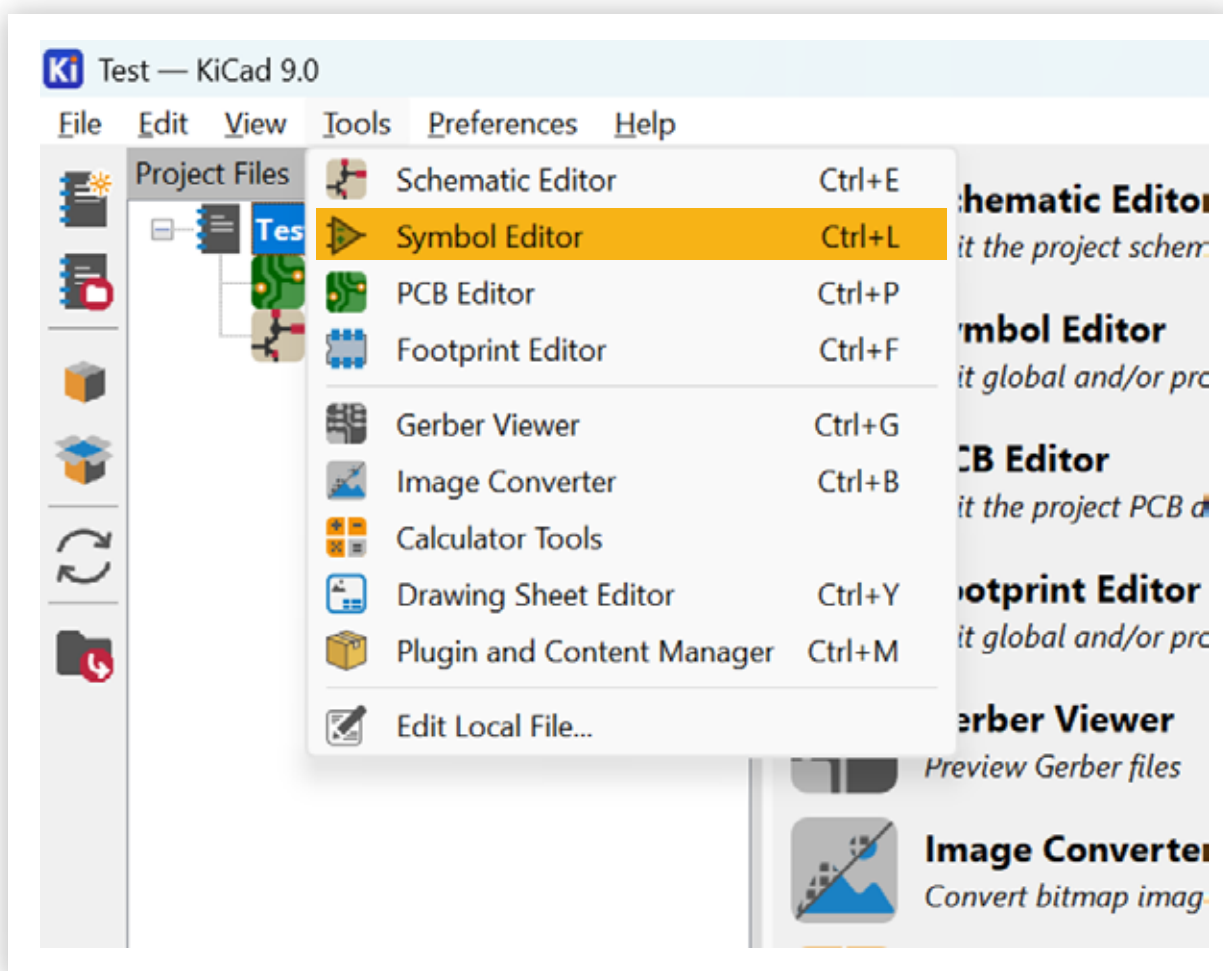
Creating folders for new symbols and footprint libraries will make it easier to copy libraries to other projects and computers.



Opening a new directory in the file explorer

1. Right-click on **(project name).kicad_pro** icon (  in the Project KiCad 9 window).
2. Select **Open Directory** in File Explorer.
3. Right-click and select **New Folder**.
4. Rename the new folder, symbols, and hit Enter.
5. Repeat steps 3 and 4, this time naming the new folder 'Footprints'.
6. Close the file explorer window.

2.4 Creating symbol libraries



Accessing Symbol Editor option

1. Open the symbol editor by selecting **Tools > Symbol Editor** (Ctrl+L). A list of symbol types will be displayed



Tip

To view the list of component types and names in the Symbol Editor library, select **View > Symbol Library Browser**. This browser window includes two columns: component types on the left and component names on the right. The component type, **Device**, can be particularly useful. Close this window after viewing.

2. To hide symbol types (optional), select **Preferences > Manage Symbol Libraries**. The Symbol Library, Global Libraries window will be displayed.

Scroll through the list of symbol types and locate the ones you want to hide in the symbol editor library window. Select the Active checkbox to deactivate or hide those symbol types you do not want to be displayed.

To close the symbol and global libraries. Select **OK**.

3. To create a new library, select **File > New Library > Project**, and then click OK. Choose one of the following options from the pop-up window:
Global: All projects will see these additions.
Project: Limits the library to your project; however, it makes copying this library to other projects and computers easier.
4. The file explorer window will be displayed. Navigate to your project and select the symbols folder. This forces KiCad to place your new library in the symbols folder that you created.
5. Rename the project file name (optional) from **New_Library.kicad_sym** to something else. Do not rename or delete the **.kicad_sym** portion of the file name.
6. Select **Save**.

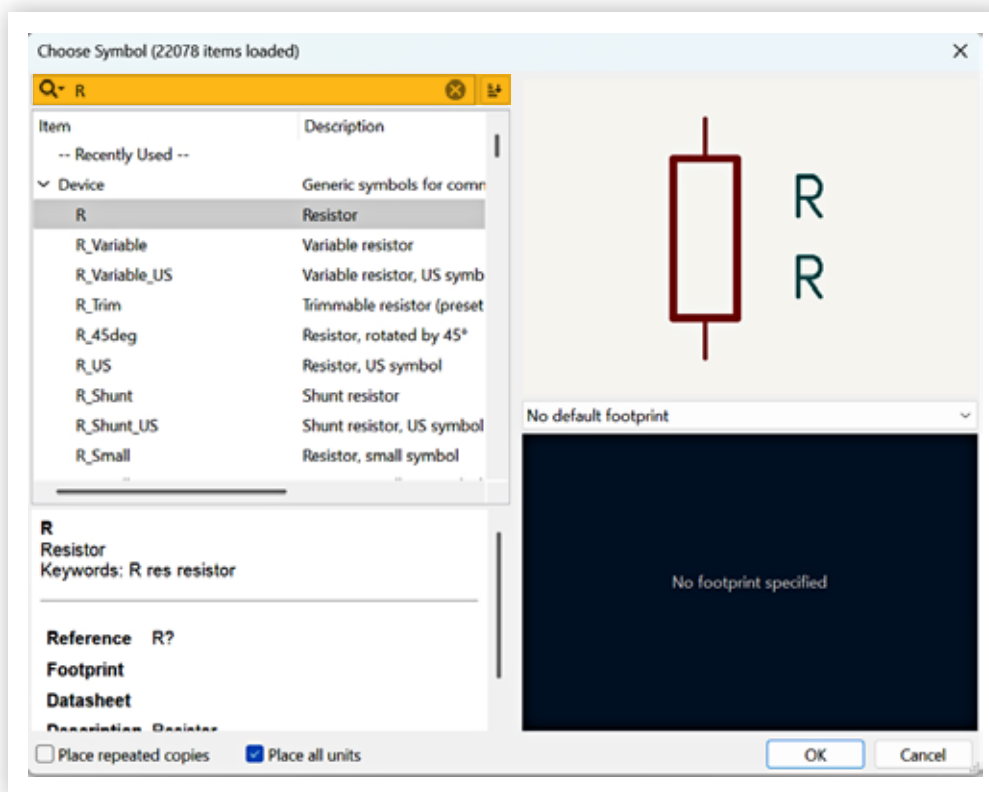


Tip

Now may be a good time to create additional libraries. Sometimes, it is useful to create additional libraries, one for all your resistors, one for all your capacitors, etc. To create additional libraries, repeat steps 3 and 6 above.

7. Right-click on your new library and select Pin Library. The pinned libraries should appear at the top of the list, marked with a "*" in front of them.
 - The new library or libraries may be buried in the list of symbols.
 - To solve this problem, scroll and find your new library or libraries.
 - Right-click on the library and select **Pin Library**.
 - Repeat this step till all your libraries are pinned to the top of the list.

2.5 Populating symbol libraries



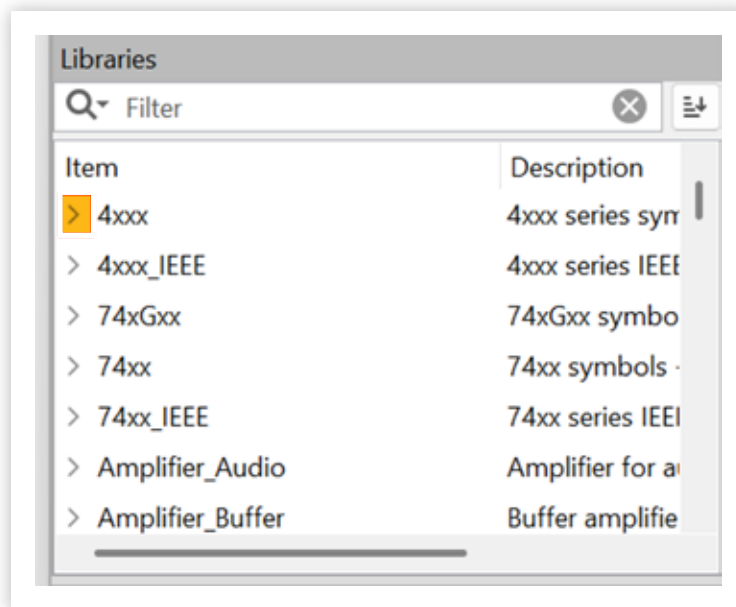
Filtering symbol in the library

1. To find a symbol, type the component's name in the **Filter** box. As you type, it shows a list of items, e.g., R for resistors or 2n to display the list of 2n**** type transistors, etc.

You could also type 'transi' to find all transistors. The filter searches through the entire symbol editor and finds all entries that include your typed entry.

The list of component types can be opened by clicking on ">".

2. Right-click on the desired component.
3. Select **Save Copy As**. The **Save Symbol As** window will now be displayed.



Drop-down option for symbol selection

4. Scroll to the top of the list and select the library where you want to deposit this symbol.
5. Select **OK**, and the symbol will be added to that library.

Tip

I recommend that you do not populate a library with more than one of each type of symbol needed for your project. For example, 2 resistors with the same value, e.g., 580k 1/4W. It's best to create additional copies of these symbols later when creating your schematic.

6. Repeat steps 2 to 5 above until you have populated the library.
7. Delete any entry in the **Filter** to display the entire Symbol Library list.
8. To delete an item from a custom library, right-click on it, select **Delete Symbol** > **Yes**.
9. Select **File** > **Save** to save the library. Unsaved libraries are displayed in bold font.
10. Close the **Symbol Editor** window.

2.6 Creating a footprint library and folder

The steps used to create and populate the footprint library are similar to those used for the symbol library.

1. Open the footprint editor by clicking on **Tools** > **Footprint Editor** (or Ctrl+F). It displays a list of components. Many of these component names end with _SMD or _THT.
 - **SMD:** These are typically surface-mount components, including electrolytic (Elec) and trimmer capacitors. Many of these components have their size indicated.
 - **THT:** These are typically through-hole types, including axial and radial mount capacitors and resistors. THT types also include diodes and other types of components. Many of these components are available in various sizes.



To view the list of component types and names in the footprint editor library, select **View** > **Footprint Library Browser**. This browser window includes two columns: component types on the left and component names on the right. Close this window after viewing.

2. To hide footprint types (optional), select **Preferences > Manage Footprint Libraries**. The **Footprint Libraries** window will be displayed.
 - Under **Global Library** tab, scroll through the list of symbol types and locate the ones you want to hide in the footprint editor library window.
 - Select the **Active** check box to deactivate or hide the footprint types you do not want to be displayed. Select **OK** to close the **Footprint Library** window.
3. To create a new library, select **File > New Library > Project**, and then click **OK**.
 - **Global:** All Projects will see these additions.
 - **Project:** Limits the library to your project; however, it makes copying this library to other projects and computers easier.
4. The file explorer window is displayed. Navigate to your project and select the footprint folder. This ensures that KiCad places your new library in the footprints folder you created.
5. Select the footprint. Rename the project File name (optional) from **Library.pretty** to something else. Do not rename or delete the ".pretty" portion of the name.
6. Select **Save**.



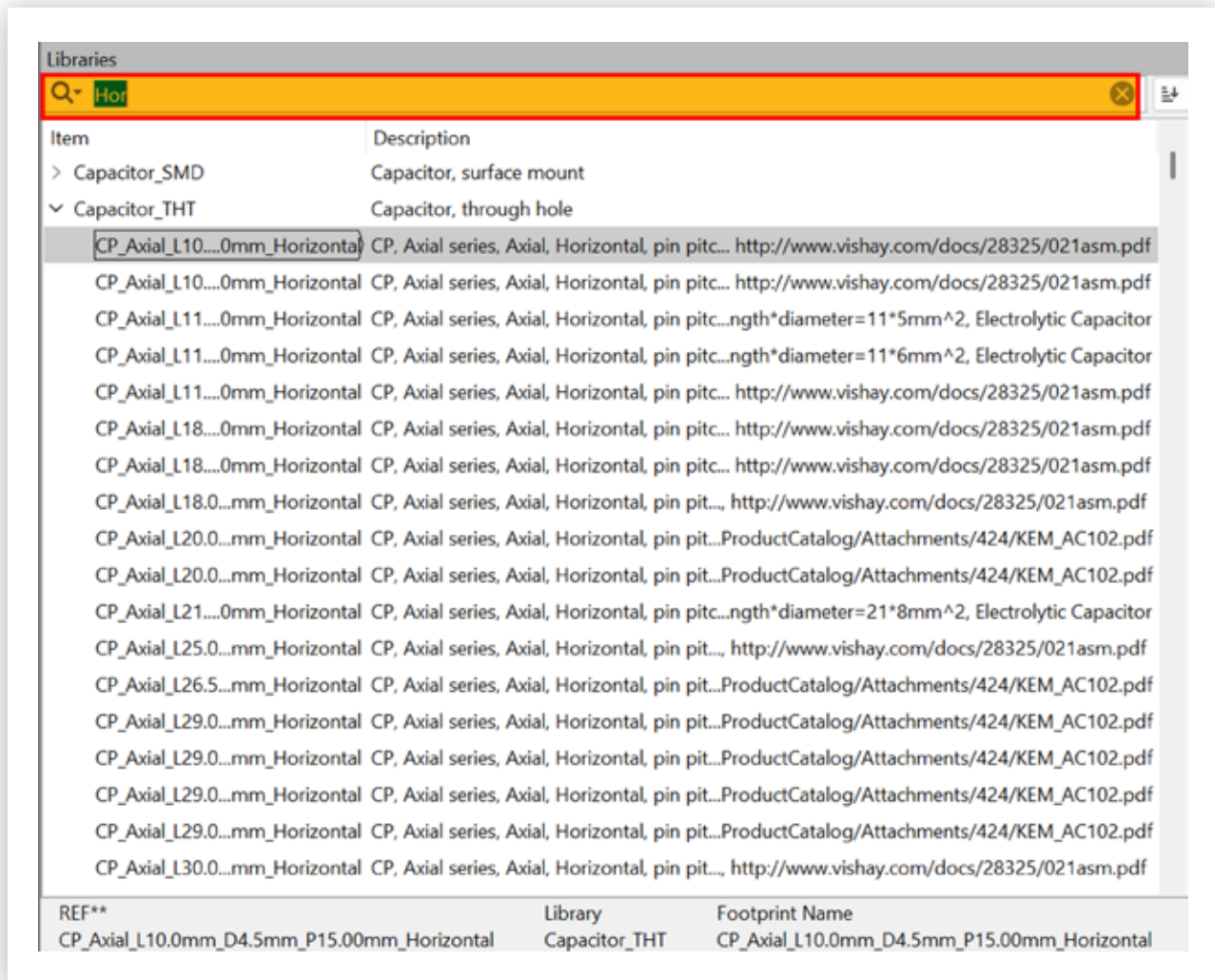
Tip

Now may be a good time to create additional libraries. Sometimes, it is useful to create separate libraries for specific components, such as resistors and capacitors. To create additional libraries, repeat steps 3 and 6.

7. Right-click on your new library and select **Pin Library**. The pinned libraries should appear at the top of the list, marked with a "*" in front of them.
 - The new library or libraries may be buried in the list of symbols.

- To solve this problem, scroll and find your new library or libraries.
- Right-click on the library and select **Pin Library**.
- Repeat this step till all your libraries are **Pinned** to the top of the list.

2.7 Populating footprint libraries



Filtering symbol footprints in the library

1. To find a footprint in a library, type the component's name in the **Filter** box. As you type, the list will narrow down to show only the items you are looking for, e.g., "cap" for capacitors.

You can also type 'hor' to find all components that include **Horizontal** in their names. The filter searches through the footprint editor and finds all entries that include your typed entry. The list of components can be opened by clicking on ">" (as shown in section 2.5).

 Tip

You may want to rename this component now. Typically, I rename resistors to include their value, e.g. 240k 1/4W. I will do the same for Capacitors and include El (Electrolytic) or Tn (Tantalum) when appropriate.

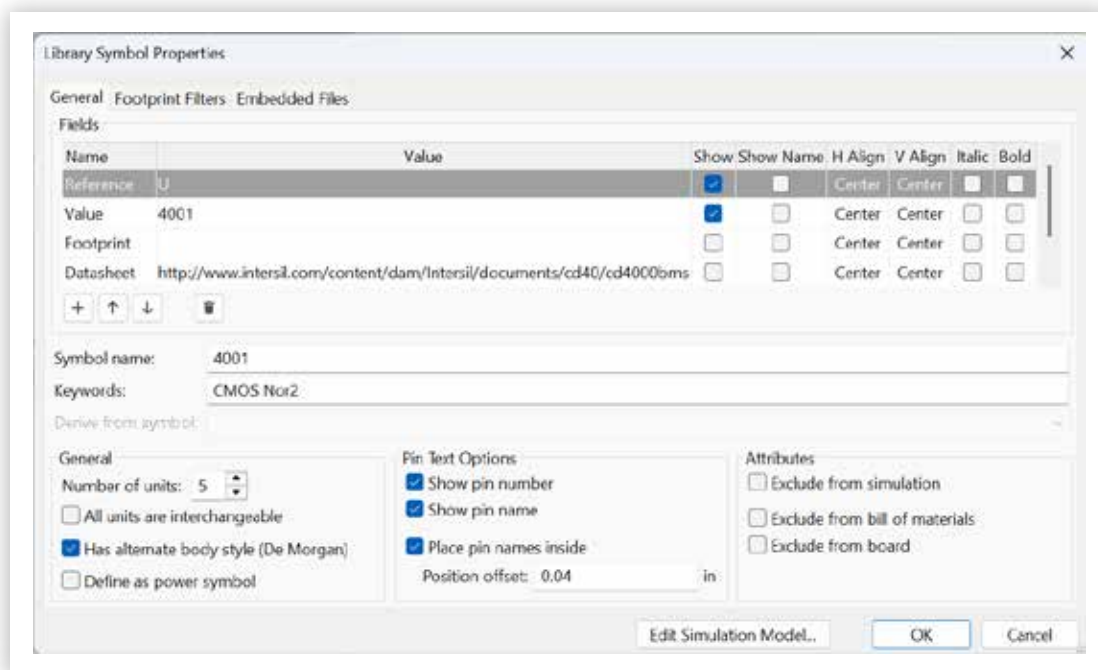
2. Scroll to the top of the list and select the library where you want to deposit this symbol.
3. Select **OK**, and the symbol will be added to that library.

 Tip

I recommend that you do not populate a library with more than one of each type of symbol needed for your project. For example, 2 resistors with the same value and/or size. You will create additional copies of these footprints later when creating your schematic.

4. Delete any entry in the **Filter** to display the entire symbol Library list.
5. Select **File > Save** for each unsaved library. Unsaved libraries will be displayed in a bold font.
6. Close the symbol editor window.

2.8 Linking footprints to symbols



Configuring symbol properties

1. Click on **Tools > Footprint Editor** (Ctrl+L).
2. Open your library. It should be at the top of the list, marked with a "*" in front of it. If you have more than one, select one to open.

3. Select **File > Symbol Properties**.

The **Library Symbol Properties** window includes 5 rows along with a header row:

- Reference
- Value
- Footprint
- Datasheet
- Description

In each row, there are 8 columns:

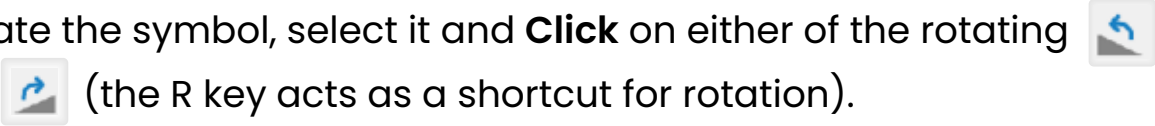
- Name
- Value
- Show
- Show Name
- H Align
- V Align
- Italic
- Bold

For this step, you will only be concerned with the Value and Show columns. To display the value on the symbol, you need to select the respective checkbox.

- **Reference:** The value field is prefilled. Do not change it or add a number to it. You may have to append numbers to this reference value as you draw your schematic. The show button is typically checked.
 - **Value:** Here, you can see the name of the symbol. The show is typically checked (more later).
 - **Footprint:** If you click in this value field, you will see the **Footprint chooser** icon  at the right end of the box.
 - **Datasheet:** You can enter the web page URL for this component's datasheet.
 - **Description:** The value box is populated with the name you assigned, but it can be changed.
4. Click the **Footprint Chooser** icon  at the right end of the box to display the **Footprint Chooser**.
 5. Open your library.
 6. Click on the footprint you want to be linked to your symbol. A footprint may be linked to more than one symbol. To the right of the documentation, a default web page for this datasheet may be displayed. If you wish to use it, copy it; otherwise, ignore it.
 7. Click on **OK**.
 8. If you want to use the copied datasheet web page, paste it in. If you want a different web page, enter it.
 9. Symbol names and keywords are usually unchanged.
 10. All other boxes are normally left unchanged.
 11. Click **OK**.

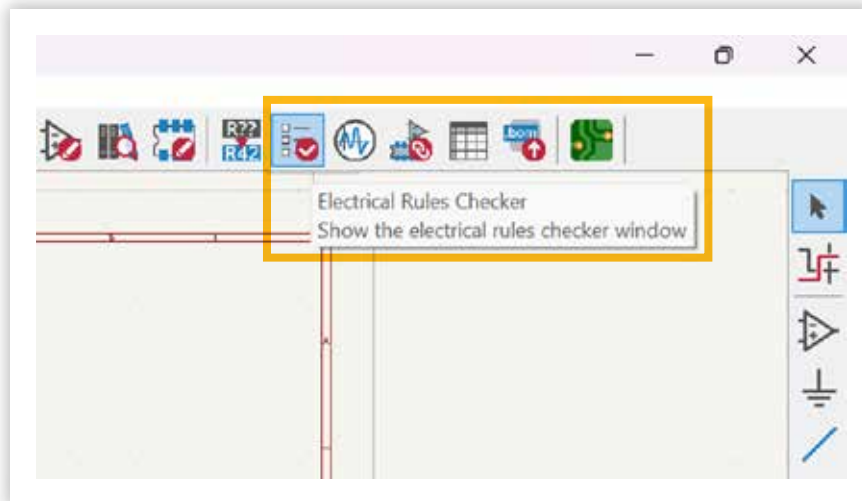
12. Repeat steps 3 through 11 above until you have filled in the properties for all symbols in your library.
13. You are now ready to create a schematic.
14. Click **Save**.

3.1 Placing symbols in the schematic

1. Open the **Schematic Editor** (Ctrl+E).
2. Select **Place** to populate the symbol.
3. Open your **Symbol Library**.
4. Scroll and find the symbol you want to place in your schematic.
5. Double-click on the symbol.
6. Position the mouse crosshairs over the location where you want the symbol to be placed.
7. Click (either the right or left mouse button) to place the symbol.
8. To rotate the symbol, select it and **Click** on either of the rotating icons  (the R key acts as a shortcut for rotation).
9. If the **Properties** window is not displayed, then select the symbol.
10. Now is also a good time to make extra copies of this symbol if you need them.
11. **Right-click** > **Copy** (Ctrl+C) > **Paste** (Ctrl+V).
12. Repeat this last step until you have all the copies of this symbol you need in the schematic.
13. Press **Esc** on your keyboard when you are done.

14. Select the symbols and verify that the **Exclude from Board** checkbox is not checked. If checked, remove the check. Repeat for all symbols.
15. Drag each symbol to its desired location in the schematic.
16. Press the **W** key or select **Place** > **Wire** from the top menu.
17. Move your cursor to a pin on a symbol (you'll see a small circle or square indicating a valid connection point).
18. Click once to start the wire, drag to the next symbol's pin, and click again to complete the connection.
19. You can right-click while drawing to add a corner or bend in the wire path.
20. To end the wire without connecting to another point, press Esc or double-click.
21. Netlabels help you organize the schematic and connect signals without drawing long wires. Press L or go to **Place** → **Net Label**, then click near the wire you want to name.
22. Enter a unique net name and press **Enter**.
23. Any other wire with the same net label will be electrically connected, even without a drawn wire.

3.2 Running the electrical rules checker (ERC)



Electrical rule checker option in KiCad 9

1. Select the **Electrical Rules Checker**  in the top Toolbar.
2. Click **Run ERC**.
3. Fix any errors and/or warnings that might be displayed.

Tip

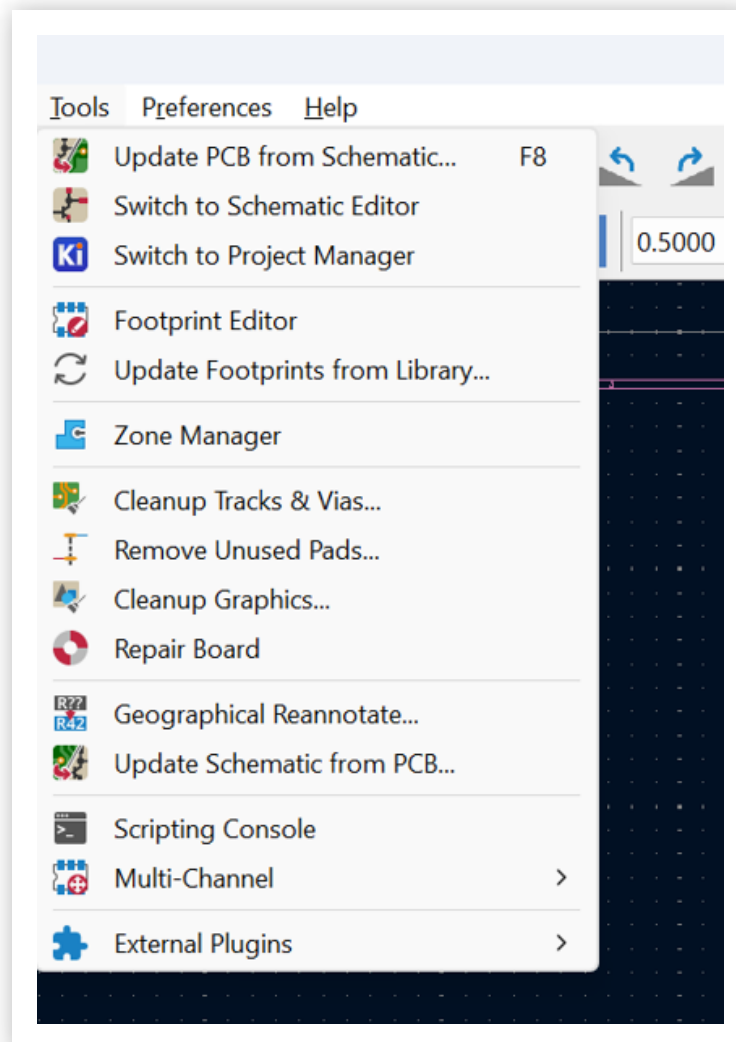
A Google search can be very helpful in identifying and solving problems. KiCad's documents and forums are loaded with great suggestions. If you need to fix Errors and/or Warnings, select **Delete All Markers** before retesting.

4. Select **Close** when you are done.
5. Click on **Save**.

Creating the PCB layout

04

4.1 Updating the PCB from the schematic



Update PCB from the schematics option

1. Select **Tools** > **Update PCB from Schematic**.
2. After a few moments, the **Update PCB from Schematic** window will display the changes to be applied.
3. Review the list of symbols to be imported. If one or more footprints are missing, these could be the reasons:

- Some symbols have an option to be excluded from the board in the symbol editor (see section 3.1, step 14).
 - Sometimes, this can be the result of duplicating symbols through copy and paste instead of using the **Duplicate** option in the symbol editor.
4. If everything looks good, click **Update PCB** to complete the operation.
 5. In the **Update PCB from the Schematic** window, 0 Errors and 0 Warnings should be displayed.
 6. Click on **Close**.
 7. All your imported footprints will be grouped in a single, compact cluster.
 8. Again, check if all the symbols have been copied from the schematic to the **PCB Editor**. If not, delete them and return to the **Placing Symbols into your Schematic** section.
 9. Before you start dragging footprints around the PCB editor, you need to assign an edge for the PCB. These will be the physical dimensions of your printed board.
 10. Select **Edge.Cuts** in the **Appearance** panel on the right-hand side.
 11. Select the **Draw Rectangle**  or **Draw Line**  tool on the right side of the PCB Editor. With the mouse, draw your rectangle or line in the PCB editor to define your board's dimensions.
 12. Now that you have defined your PCB in the steps above and successfully imported all your footprints, making your work easier, separate all the footprints by dragging them to new locations.



Tip

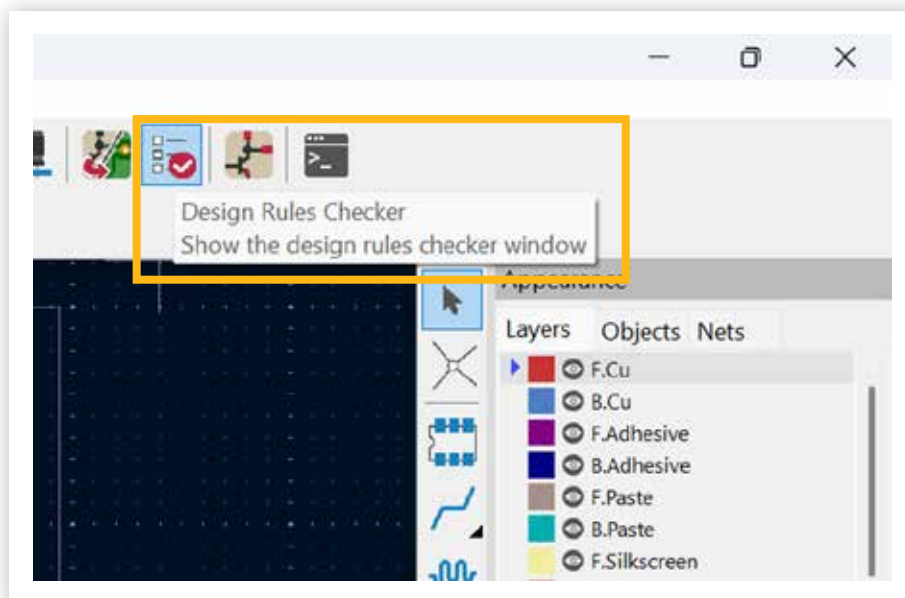
While dragging items, they may sometimes snap to locations that the software prefers. You can turn off this **snapping** function by pressing the Shift key on the keyboard while dragging things.

13. As you drag these footprints, you will notice thin lines between the footprints (components). Some call these a **Rats Nest**. I call them **Spider webs**. In any case, they represent the connections between Footprints brought forward from your schematic.
14. They can help guide the placement of footprints on your PCB and can be ignored. From what I can tell, each one of these lines represents a software-generated net.
15. Look at the **Appearance Layers** list on the right portion of the window. There is a small triangle in front of one of the Layer options. Check that all components have been placed on the correct layer. They should be on the copper layer. Failure to do this could result in your components being located on the wrong PCB layer.
16. If the incorrect layer is selected,
 - Select the group of components.
 - Go to **Edit > Cut** (Ctrl+X).
 - Choose  **F.Silkscreen layer.**
 - Select **Edit > Paste** (Ctrl+V).
17. Now, drag the components to the desired location in the **PCB Editor** window.
18. Remember to use the  and  icons in the top toolbar to rotate symbols.
19. Set up PCB design rules before routing.
20. Go to **File > Board Setup** or press **Ctrl+Shift** to open the **Board Setup window**. In the left pane, expand the **Design Rules** section.
21. Under **Net Classes**, define or edit rules such as:
 - Track width
 - Clearance between tracks
 - Via size and hole diameter
 - Clearance between copper features and board edge

22. Add custom net classes if your design includes high-speed signals, differential pairs, or power nets. Assign the appropriate rules per net class.
23. Under **Constraints**, set global limits such as:
- Minimum copper-to-edge clearance
 - Minimum track/via dimensions
 - Solder mask expansion
 - Silkscreen clearance from pads
24. Confirm and apply your settings by clicking **OK** or **Apply**, then close the **Board Setup** window.
25. These rules will now be enforced during routing and design rule check (DRC), helping to catch issues early in the layout process.
26. It's time to add your route tracks or circuit traces.
- First, make sure that the correct copper layer has been selected. In a single or two-layer board, they will be **F.Cu** and **B.Cu** for the front copper layer and the rear copper layer.
 - To add a trace or track, click on the **Route Single Track**  icon in the Toolbar on the right.
 - Bring the mouse pointer to the location where you want the trace to begin.
 - Traces should always be drawn from one footprint pad  to another footprint pad or trace.
 - Click one time to start your trace.
 - Additional single clicks can be used when you need your trace to include a 90° bend.
 - Double-click to end a trace. As you draw each trace, the PCB Editor will faintly display a net name the software has assigned to that trace.
 - Repeat steps b through f above for each additional trace required.
 - Use the **Delete** key on your keyboard to delete errors.
 - Press the **Esc** key on your keyboard to exit the **Route Single Track function**.

27. Select the **T** key on the Toolbar on the right to add text to your silkscreen.
28. If you are satisfied with all your work, proceed to the next step. You can always return to the **PCB Editor** window and make additional changes.
29. Select **File and Save** to save your work.

4.2 Running the design rule checker (DRC)



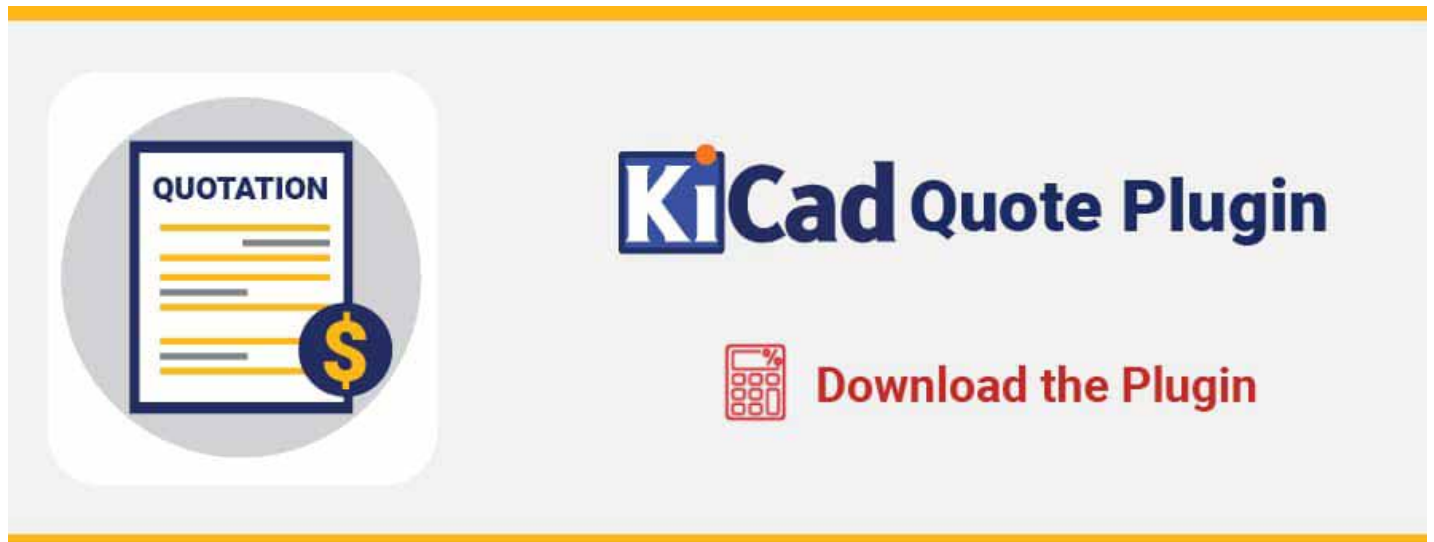
Design rule checker option in KiCad

Once again, it's time to check your work.

1. Click on the **Design Rules Checker**  icon in the top Toolbar.
2. Click on **Run DCR**.
3. After the Electrical Rules Checker has completed checking your footprints and tracks, it will display all errors and warnings found.
4. To display only the errors or warnings, uncheck the errors or warnings checkbox.
5. A Google search can be very helpful in identifying and solving problems. KiCad's documents and forums are loaded with great suggestions.

6. Run the DRC as many times as needed until you have eliminated all errors and warnings.
7. Select **Close** when you are done.

Install Sierra Circuits' KiCad plugin to get an instant quote:

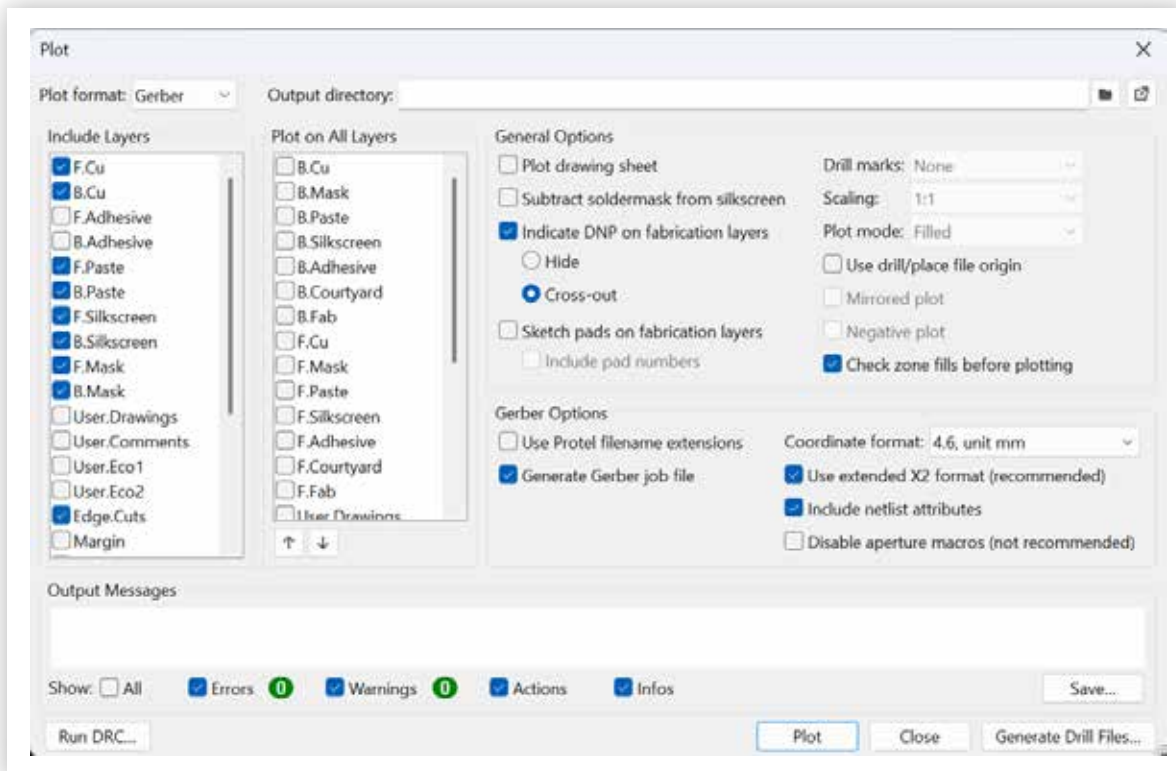


Features:

1. **Seamless integration:** The plugin integrates directly within the KiCad design environment, allowing you to obtain instant PCB fabrication and assembly quotes without leaving the software interface.
2. **Automated design analysis:** It automatically reads your PCB design files to extract essential parameters such as board dimensions, layer count, and other specifications, streamlining the quoting process.
3. **User-friendly interface:** The plugin offers a simple and intuitive interface, making it easy for users to review and adjust specifications like board thickness, surface finish, quantity, and turnaround time.
4. **Flexible ordering options:** Users can choose between PCB fabrication only or opt for complete turnkey services, including assembly and component sourcing.
5. **Real-time quoting:** By eliminating the need to export design files and manually input data into external quoting tools, the plugin provides real-time quotes, reducing the potential for errors and saving valuable time.

Downloading production files

05



Configuring plot properties for Gerber download

5.1 Creating Gerber files

Gerber files are the standard format used to describe the design of a PCB. They include information about all the individual board layers, including solder masks and drill data. Your fabricator will use these files to create your circuit board.

1. Select **File > Plot**.
2. Select **Gerber** in the Plot Format dropdown box.
3. **Output directory:** Use the **Search** tool to find the directory where you want these files to be stored. I recommend creating an additional folder beside the symbols and footprint folders.

4. **Include layers:** The default ones are normally correct. Check that all the important layers are included (*.Cu, *.Paste, *.Silkscreen, *.Mask, Edge.Cuts).*. Adhesives are only necessary if components are to be glued to the board during assembly.
5. **General options:** These recommended options include **Indicate DPN on fabrication layers** (Cross-out) and checking zone fills before plotting.
6. **Gerber Options: General Gerber job file, Coordinate format** (typically 4.6-unit mm) use extended X2 format (depends on the fab house), and Include netlist attributes.
7. Select **Save**.
8. Click on **Plot**.
9. Choose **Generate Drill Files**.
10. Leave the default settings.
11. Select **Generate Report File > Generate**.
12. Close all windows when you are done.
13. All Gerber files have a ".gbr" extension. Other important files include the -drl. rpt, *-gmrjob, *-NPTH.drl and *-PTH.drl files.
14. Close the **Plot** window.
15. Before sending these files to a manufacturer, zip them up.
16. Close the **PCB Editor** window.
17. Select **Save**.
18. Close the Schematic Editor window if it is still open. If prompted, select **Save**.

5.2 Generating a bill of materials

1. Open the **Schematic Editor** (Ctrl+E).
2. Select **Tools > Generate Bill of Materials**.
3. To edit the appearance of the output file, select **Edit**.
4. Check and uncheck the **Show** checkboxes and view the results in the area to the right.
5. Repeat this step for the **Group By** check boxes.
6. In the area to the right, you can edit the order by dragging any column right or left.
7. Select **Export**.
8. Here, you can change the default field, string, and reference delimiters. You can also add range delimiters and include options for keeping tabs and line breaks.
9. Select **Export**. The output destination is displayed in an Information box. Select **OK**.
10. In File Explorer, look for a file with a .csv extension in your project folder. If you have Excel or any program that can read CSV files, clicking on this file will open your BOM.



Tip

My experience with the files in Excel is that the information in the value column may not always come across properly. Excel does not recognize Ω , so it is replaced with $\hat{\text{I}}\text{C}$. I use the find and replace function (Ctrl+H) to fix this.

11. Select **File > Save As**.
12. Double-click on **Recent**.

13. In the **Save As** window, select the file name to be saved in the name list.
14. Select **Save > Yes** to **Do you want to replace it**.
15. Close the Project – KiCad window to exit the program.
16. Congratulations, you have completed a PCB project.

Other Resources on KiCad Design Guide



For further learning, check out:

- [Sierra Circuits' KiCad Design Guide](#)
- [KiCad's Official Documentation](#)
- [KiCad 7 & 8 Beginners Guide to Personal Symbol and Footprint Libraries](#)
- [Getting Started in KiCad 9](#)
- [KiCad 9.0 Reference Manual](#)

About Sierra Circuits



Sierra Circuits has been serving PCB designers and engineers with the latest technologies since 1986 and has worked with over 20,000 customers since then. We specialize in manufacturing, assembly, and HDI technology. We handle all aspects of circuit board production.

We provide our customers with DFM and DFA support, which helps produce unprecedented board quality with high reliability. By choosing us, you can eliminate miscommunication between multiple vendors and delays since we provide a single point of support.

Talk to our experts

Sierra Circuits helps PCB designers plan it right!

Our engineering staff has been trained on controlled impedance and can analyze the design from a holistic point of view.

Our engineering support and our stack-up team provide valuable suggestions with their knowledge of controlled impedance for high-speed designs, analog/digital, high-density board manufacturing design rules, and design for assembly guidelines. Upload your data and receive a free consultation and review of your design. Services include system-level design, schematic capture, circuit board layout, and PCB/PCBA DFM.



Our facility

Why Sierra Circuits?

Customer-oriented approach: We collaborate closely with you and tailor our solutions to meet your unique design requirements.

Rapid high-tech innovation: We work in a fast-paced environment and implement cutting-edge technologies to bring your design ideas to life.

Integrated systems: Integrated turnkey world-class team that includes design, engineering, manufacturing, support services, and supply chain that engages at the beginning of all of our partnerships.

Focus ready: We focus on complex products for defense and aerospace that require high reliability and repeatability to support mission-critical systems on time.

Sierra Circuits’ PCB manufacturing capabilities overview

Rigid PCBs

PCB highlights	Fabrication highlights	PCB classification	Quality management systems
Up to 30 layers	Routed arrays	IPC 6012, class 1, 2 and 3	AS9100D
Board thickness: .005" - .250"	Edge plating	MIL-PRF-55110	ISO 9001:2015
Max panel size: 21" x 29"	Bevels	MIL-PRF-31032/1	ISO 13485:2016
Min trace and space: .002"	Heat sinks	MIL-PRF-31032/2	
Solder mask feature tolerance: .001"			
Blind and buried vias			

HDI and microelectronics

High tech in small form	Features	Certified for your industry
Special stack-up help / NPI review	Down to 1.5-mil trace space, 2-mil holes	Mil-spec
Up to 30 layers	Blind vias, buried vias, and other microvia techniques	Automotive
Files evaluated by engineers	Up to 5 sequential lamination cycles	Aerospace and defense
Quote sent by email within a few hours	Laser direct imaging	Medical devices
	via-in-pad technology	

Flex and rigid-flex PCBs

Surface finishes	Standard flex materials	Fabrication highlights	PCB classification
HASL	0.5 mil to 5 mil thick polyimide material	Routed array	IPC 6012, Class 1, 2, and 3
Lead-free HASL		V score	
OSP (Shikoku F2 and Entek)	1 mil to 5 mil thick copper clad base material	Edge plating	ISO:9001:2015
ENIG	Flame retardant laminate	Edge castellation	ISO 13485: 2015
Immersion silver		Countersink	
Immersion white tin	UL and RoHS compliant	Bevel	
Tin nickel		Milling	
Selective gold			

Aerospace and defense PCBs

Capabilities	Documents we provide	Ideal for
Layer count: Up to 30	Certificate of conformance	DOD product designers developing systems for mission critical functions
Minimum board thickness:10 mil	Material specifications	
Thickness tolerance: 10%	Reflow profile copy (included with first article)	Contract manufacturers who need fast, predictable lead times
Minimum trace and space width: 3 mil	Photo requirements	
Minimum drill-to-copper clearance: 6 mil	First article inspection report	Purchasing teams who require pricing and supply-chain transparency
Maximum panel size: 21" x 29"	Record of calibrated tools used during manufacturing	
Controlled impedance	FPT, AOI, ionic cleanliness report	Aerospace engineers looking to shore-up their supply chain troubles
Stacked vias		

RF and microwave PCBs

Capabilities	Certifications
Operating frequency: 10 MHz to 30 GHz	PC Manufacturers Qualification Profile (MQP)
Sodium etch hole treatment process	IPC-A-600
Full-body gold plating up to 120u"	IPC-6012 standards
Soft gold without nickel or ENIG surface finishes for enhanced board protection	ISO 9001:2015
	NADCAP
	ITAR registered
Design assistance	RoHS/REACH standards

Sierra Circuits’ PCB assembly services

We offer a diverse range of PCB assembly services tailored to meet your unique requirements. Whether you need a full turnkey assembly, consigned or partially consigned assembly, we've got you covered.

Full turnkey PCB	Partially consigned assembly	Consigned assembly	Assembly capabilities
Design review	Some parts provided by the designer	Components are provided by the PCB designer	BGA, micro-BGA, QFN,CSP, lead less devices up to 0.35mm pitch, press fit components
Bare board fabrication	We will source the rest	Sierra Circuits will fully assemble your PCBs and ship them to you.	DFA
Component procurement	Supply chain resilient	Quick turn times	RoHS, leaded, indium, clean and no clean chemistry
Testing	Quick turnaround time	Fully transparent pricing	Component sizes: 0201, 01005, 08004
Quick turnaround time			Paste in-hole

To help you overcome the challenges involved in component procurement, Sierra Circuits has introduced customer-owned inventory service (COIN). It's a personalized component sourcing and stocking program. We procure and store parts for your PCBAs. This can help you significantly reduce your product development time.

Sierra Circuits' PCB design services

Whether you're beginning with a new design or enhancing an existing one, our design experts provide the assistance you need. Our PCB design services include stack-up design, signal integrity analysis, circuit board layout optimization, schematic capture, and reverse engineering.

Our capabilities include:

- High-speed digital designs, analog designs, mixed designs, power designs, and RF designs
- Controlled impedance with a tolerance of $\pm 5\%$
- Mechanical-constrained boards and enclosures
- Flex, rigid, and rigid-flex designs
- PCB designs for mobile phones
- HDI designs with via-in-pad, blind via, buried via, microvias, via stacking, and staggering technology
- Fine-pitch BGAs of 0.4 mm and 0.5 mm
- Designs with high-pin count full matrix BGAs 0.4 mm pitch, PoP package, with 2 mil/ 2 mil tracks/spacings
- Design for ROHS compliance
- Optimum impedance calculation for single-ended, differential, and coplanar-waveguide models

Advanced PCB design tools

From material selection to impedance calculation, streamline your design process by using our comprehensive suite of cutting-edge PCB design tools. These web applications enable you to expedite your design process and make the right decisions to ensure signal integrity and reliability in your circuit board.

Sierra Circuits PCB certification and registration

IPC certified for quality standards in PCB fabrication and assembly

- IPC Manufacturers Qualification Profile (MQP) for Sierra Circuits, Sunnyvale, California
- IPC-A-600 inspectors

ISO certified for military, aerospace, and medical device PCB fabrication and assembly

- ISO 9001:2015 certificate
- AS9100D – military and aerospace
- ISO 13485:2016 certificate – medical devices

Mil-Spec certified and DLA approved

- Acc 2020 CVI VQE 035113
- JCP certification approval letter

ITAR Registered

- ITAR registration letter

Underwriters Laboratory certified

- Flex and rigid flex PCBs
- Rigid PCBs

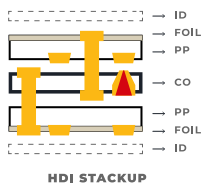
Exceptional material compliance and reporting

- RoHS / REACH compliance material composition declaration
- Conflict minerals reporting template

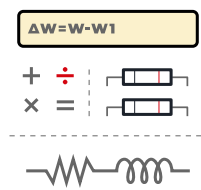
Certified minority supplier by NMSDC

- Minority business certification

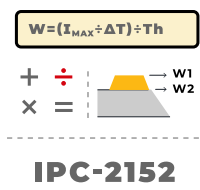
The Designer's Toolkit



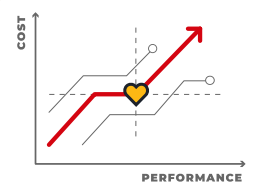
Stackup Designer



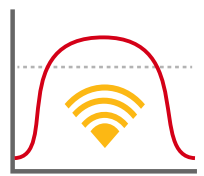
Impedance Calculator



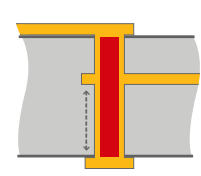
IPC-2152
Trace Width,
Current Capacity and
Temperature Rise
Calculator



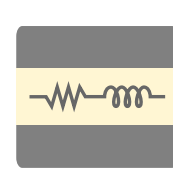
Material Selector



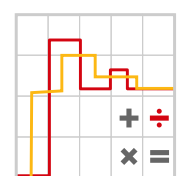
Bandwidth, Rise Time
and Critical Length
Calculator



Maximum Via
Stub Length
Calculator



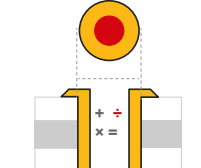
Signal Layer Estimator



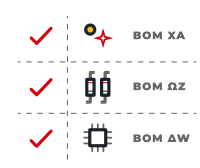
Transmission Line
Reflection Calculator



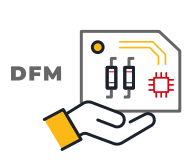
RLC Resonant
Frequency &
Impedance Calculator



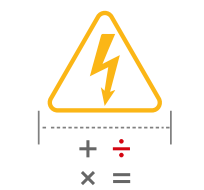
Via Current
Capacity and
Temperature Rise
Calculator



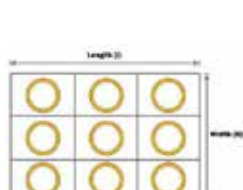
BOM Checker



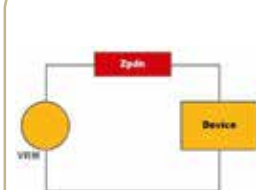
Better DFM



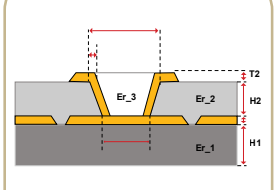
Conductor Spacing
and Voltage Calculator



Via Thermal
Resistance Calculator



Power Distribution
Network Analyzer



Via Impedance
Calculator

Try Tools



Sierra Circuits, Inc.

1108 West Evelyn Avenue, Sunnyvale, CA 94086, USA

This design guide offers step-by-step directions for using KiCad,
a powerful open-source PCB design tool.

From project setup and schematic creation to layout, rule checks, and Gerber downloads,
the guide covers the complete PCB design workflow in KiCad. Whether you're starting a
new design or building upon an existing project, this manual is structured to help you
confidently navigate every phase.